

Lec 50 Classifier Guided Diffusion Model

Lec 50 Classifier Guided Diffusion Model

Lecture	50 / 63 + intro
Duration	37:35
Captions	English (auto-generated)
Words	5,109
Video	tDFtVtmwuxo
Playlist	Fundamentals of Generative AI and LLMs

Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone, welcome back to the lecture series on fundamentals of generative AI and large language models. In our previous session, I have discussed about the unit that is used for denoising in diffusion models. In today's session, uh we will discuss about classifier guided diffusion model. So the contents of today's session goes like the motivation behind classifier guided diffusion models. What exactly is classifier guided diffusion? The mathematical formulation of it, guidance scale, sampling equation and the limitations. Okay. So we will start with the motivations.

[00:45] Now we know in standard DDPM the model learns the data distribution and it generates realistic images but it has no control over which kind of image it will generate which class of image it will generate. So when I'm taking a noise and my uh model is trained to generate images from that you know noisy uh version of an image. It will generate something but there is no guarantee that the generated images will be either of class dog or cat class or flower class or any such specific class. Okay. So if the requirement is such that I need to generate only images of dog, how do I do that one? Okay. So that was the question that scientists you know uh found and they tried to find out an answer using this classifier

[01:46] guided diffusion. So as I [clears throat] was telling so in normal diffusion or DDPM what we have we have the diffusion model here and then we input a noise and it generates some images. Now those images can be anything. It can be cat, it can be a car, it can be dog or it can be a bird. There is no control on what the diffusion model generates. Now can we force the model to generate a dog instead of a random image? So in that case what happens you have a noise that gets uh as input to the diffusion model and then an image is generated from that and that image corresponds to images of dogs. Okay. Now here there is various kinds of dogs. You can make it very specific about a particular breed of dog as well. Okay.

[02:48] So for this one, if we want to do a uh generation of images which are of a particular class, then we need some guidance mechanism. Okay, without a guidance mechanism, it is not possible. Without a guidance mechanism, the uh model will not know what exactly I want the model to generate. Okay. So that is where our classifier guided diffusion models came into picture. So what exactly is the core idea take the idea is like this. Take a generative model that knows how to create anything. It knows our normal diffusion model. It knows how to create an image from a noise but it doesn't know only what is the limitation. It doesn't know what to create. Okay. So it keeps on creating random images. So the core idea of classifier guided

[03:44] diffusion is take a generative model that knows how to create things and then give it a GPS or a steering wheel which it steers it towards a specific category. So the image generation is steered towards a specific category. Okay. So that is the core idea. Now how the implementation is done that is that is what we will see in this lecture. So the idea of classifier guided diffusion was first introduced in the paper diffusion models beat GANs on image synthesis by Dhari P and Nicole A and it came up in the NeurIPS 2021 conference and we will see what exactly this paper has or what is the um how is the implementation of the classifier guided diffusion happens.

[04:38] So in a com the there are various components in a classifier guided diffusion model. What are they? First is a normal diffusion model which acts as an artist and knows how to denoise an image into something realistic but it does not know which class to generate. Okay. So it acts as a normal denoising diffusion probabilistic model. This has no conditioning thing in it. And then we have a classifier which acts as a critic and which looks at the noisy image and predicts the probability of each class given a noisy image. So the classifier what is the function of the classifier?

[05:17] It looks into the noisy image that is created by the diffusion model and predicts a class and this classifier basically itself it is a pre-trained model. So it has been trained with all general uh purpose photo uh you know images like cat, dog, trees, houses you know so many uh classes of images. So when it sees a noisy image, it tries to predict which class it is belonging to. Okay. So two components are the diffusion model and the classifier.

[05:52] Okay. Now we will see how it works. So we have the artist and the critic. As I was telling the it's a you know intuition you can build and that makes the mathematical formulations of classifier guided diffusion model much more clearer to you. So the artist what it does it takes a small step to remove noise from the image. So we know the artist is nothing but your normal diffusion model. Okay denoising diffusion model or the DDPM.

[06:24] And what is the critic doing? Critic is your basically your classifier. It looks at slightly clearer image and calculates the gradient of the class. Say for example, if it is a dog class, then what is the gradient of the class label with respect to the image pixels. So it will look into each and every image pixel and then it will find out what is the gradient with respect to x_t or with respect to the noisy image. Okay. And then what it will do there will be an adjustment done based on this gradient. So the classifier gradient indicates how to modify the image to increase the probability of a specific class. So in this case we were talking about dog class. So here the classifier gradient how it will help the model. It indicates

[07:16] how to modify the image in which direction the denoising should happen so that my generated image looks more like like a dog's image. Okay. So it increases the probability of a specific class say class Y in this case Y is equal to our dog class. And this is the mathematical equation or the expression for the classifier gradient. We will look into more details of this one in our coming slides. But this you know the uh gradient is taken with respect to x_t which is your noisy image pixels or noisy image. Then $p(y|x_t)$ which actually tells you what is the probability of the class y given you have a noisy image x_t .

[08:11] And this ϕ is basically the parameters of the neural network. Okay. Now once you have this classifier gradient then there is an update. Where is the update happening? The diffusion model shifts its denoising towards the that gradient. That means diffusion model. What is the function of the diffusion model? In each and every step the diffusion model is denoising the noisy image and making a clearer version of that one. Now with the addition of the classifier gradient what happens is the diffusion model shifts its denoising direction in a particular direction so that the generated images looks more like class Y or in this case dog class. Okay.

[09:05] Now there is something called as the guidance scale. What is that one? Guidance scale basically means how much with what um uh strength you want to give the classifier gradient to the model. Okay. So it can vary from one to you know numbers higher numbers but it is seen that increasing the guidance scale makes the image more recognizable as the target class which is correct as you increase the guidance value. That means the strength with which you are adding this um class specific gradient to your unit model or to the denoising model.

[09:59] You are getting very precise or recognizable dog images but it often reduces

the variety of the images the model produces. So the diversity reduces at the cost of precision. So there is a trade-off between how much will be your uh guidance scale which is nothing but the strength of the gradient that you want to give to your denoising diffusion model. Okay. So if it is too high then you get very precise or accurate dogs images but the diversity of the images generated suppose 100 images are generated the diversity in them is reduces. So you have to find out an optimal value of S or optimal value of the guidance scale which will be a trade-off between these two. Okay.

[10:57] Now orientation of image generation towards a particular class. How that happens? I have told that when we give the gradient of the classifier to the denoising diffusion probabilistic model without any condition then the image generation is oriented towards a particular class. So we will look into more details about that one here. [snorts] So a DDPM learns a reverse transition. We all know about this equation. $P_{\theta}(x_{T-1} | x_T)$ because it is in the reverse transition.

[11:30] It starts from $x_T = \text{capital } T$ and then goes till $x =$ to uh T equal to 0 or x_{x_0} is can be represent at a as a Gaussian distribution with mean μ_{θ} which depends on x_T and T and coariance σ_{θ} which once again depends on x_T and T . So this is a normal DDPM. Here if you see there is no conditioning that is coming into picture. How do we know whether the equation is having conditioning? If there is that y term there which is the class level in whatever we have discussed so far. Okay.

[12:09] Now part of this denoising trajectory in the reverse process to move it towards a particular class and then generate image from for that particular class. Okay. So what exactly? So this is uh the reverse transition. We are aware of this equation. Now what we have to do to orient it to some particular class of image generation, it is simple. The idea is you perturb that reverse uh trajectory of denoising. And how will you perturb you will perturb it with the gradient that is coming from the classifier. Okay. And that will help in directing your image generation towards a particular class. Okay.

[12:56] So how do we start? We start with training a noisy image classifier. Okay. So you have a classifier which is trained with lot of noisy images. Okay. So you can see there is a dog jumping dog and there are various noisy versions of that one and all of them go to the classifier during the training phase and the classifier knows which class it gives the class prediction. Okay. Now $p(y | x_t)$ is the probability that the noisy image x_t belongs to class y .

[13:34] So whenever I write $P(y | x_t)$, it means that I'm talking about the probability that the noisy image x_t belongs to class y . Okay. Now what happens? So this is during the training of the classifier. So as I told the classifier needs to be trained with noisy images. If we take a classifier which is trained with clean images will it work? It will not work because during the denoising process or the reverse process

of diffusion the classifier needs the what the classifier will have is the noisy images not clear image. So as long as it is not trained with the noisy images it will not be able to find out the gradient.

[14:28] Okay, properly. Now what happens during the sampling or the inference term uh inference time. Okay, so assume your DDPM is trained. So your DDPM is trained the normal DDPM which is not having any conditional information. So it knows it has gone through forward diffusion process. It is trained and now it is trained in the reverse. it is now going through the reverse diffusion process to generate the images and you will sample from that uh distribution of images. So during sampling what happens is a noisy image is given to the network. Okay.

[15:07] Then the unit predicts the noise component epsilon theta which depends on x_t and t present in the noisy image. Okay. So we all know that in the denoising process the unit predicts the noise component that is present in the image and then that image is that noise is subtracted to get the clearer version of the image. Okay. So once the noise prediction is done for this unconditional or normal DDPM that amount of noise is removed from the image to get the clearer image. So this is known and then what is happening is the gradient from the classifier is also given as input to the unit. Okay. So we will have a look here. So you have the unit diffusion model or the unit uh model for uh denoising the diffusion uh

[16:02] you know denoising process and then you have a noisy image that is given here to the unit for denoising purpose for predicting the amount of noise that is present in the image and then you have a little clearer version of the image and together with that one you are taking the classifier and taking the gradient of the classifier with respect to x_t . What is your x_t ? x_t is the noisy version of the image. Okay. So that is going as the input. So this is your x_t . And then you take the gradient of the classifier with respect to x_t . Then these two things are added to give you a clearer image of the in the next step. Okay. So by adding this gradient of the classifier what we are doing we are steering the wheel towards generation of

[16:56] images which belongs to dog class. Okay, that's the idea. And the gradient of the classifier as I was telling is you know it's the gradient is taken with respect to $x_t \log p_{\theta}(x_t | y)$ will be the you know θ parameters of the neural network or y given x_t . Okay. So this is what we have um all discussed in a previous slide. Okay. Then we will have a look into the mathematical formulation. So what exactly the mathematics tells us or how do we interpret this entire process mathematically. Okay.

[17:40] So in the sampling uh process we would like to sample from P of x_t given y . That means what exactly it is telling the probability of observing a noisy image x_t which is conditioned on y . What is your y ? y is the class level in this case consider it as a dog class. Okay. So probability distribution of images $P(x_t | y)$ it is the probability of observing a noisy image. x_t is basically your noisy image and it is

conditioned on the class label Y . Okay.

[18:18] Now if I use base theorem here for conditional probability what we get is P of X_T given Y can be written as P of probability of Y given X_T into P of X_T divided by P of Y . Okay. So this is from base theorem which is a standard base theorem uh which is used for conditional probability. Okay. Now I have just written this expression using base theorem. Then what I do? I take log on both sides which gives me log of this term.

[18:54] Log of $p_{X_T|Y}$ equal to log p_{X_T} which is your this term. p_{X_T} comes from this term. Okay. Plus log of p_Y given X_T which is your this term minus log of p_Y which is coming from your this term. Okay. Now since p_Y is constant it doesn't depend on X_T . If I take the gradient which I will take in the next step with respect to X_T that log of p_Y the gradient of the term will become zero. Okay. Now what we are doing taking gradient with respect to X_T on both sides. Okay. So what we will do? So when we are taking the gradient of this log term. So you know gradient with respect to X_T log $p_{X_T|Y}$ equal to log of p_{X_T} okay gradient of with respect to X_T log of p_{X_T} plus this terms gradient

[20:00] which is gradient with respect to X_T log p_Y given X_T . Okay. So what we have taken here is gradient with respect to X_T log p_Y equal to I have taken zero because p_Y is constant it doesn't depend on X_T what is p_Y what is Y is the class label so this class label doesn't have any dependency on your noisy image the noisy image can come from any form But is your class label dependent on the noisy image that is coming to the model? No. That is the reason when we take the gradient with respect to X_T then log of p_Y becomes equal to zero. So we get this equation.

[20:51] Okay. After that what happens? We will see. So this is a key equation of the classifier guidance or classifier guided diffusion model. Then we will see what happens here. So I have just written once again that same equation. Okay. Now we will see this part is basically responsible for creating realistic images. Why? Because there is no there is only this term is there X_T which is basically your noisy version of the image. It doesn't have anything related to the class label Y .

[21:24] Okay. So this term is giving you or is deciding how much the image will be realistic. How much the generated image will be realistic. Okay. And the second term here is telling how much your generated image is going towards the target class. So we have two terms here to have a conditioned generation you have two parts. One is the image becoming realistic plus one the other other term is uh telling you how much the generated image is looking like the target class in this case like a dog's image. Okay.

[22:14] So this is basically a gradient of the classifier. Okay. P of Y given X_T . That means the probability that your noisy image X_T is looking like class label Y . Probability that your noisy image X_T belongs to the class label Y . Okay. Now this expression or

this term in normal unconditional DDPM itself can be written as $-\frac{1}{\sqrt{1 - \alpha_t}}$ by square root of $1 - \alpha_t$ this we have seen in our previous lecture also what is α_t ? It is equal to product of α_i for $i = 1$ to t . Okay. So this is the α_t expression. This we have used multiple times in my previous lectures and it represents the cumulative amount of original signal that remains after t forward diffusion

[23:21] steps. Okay. So it represents that one. Now this can be written as if I tell this is a score. This is written as or the diffusion score. This can be written as a constant. What is your constant? Constant is this term. And the noise prediction. Noise prediction is $\epsilon_{\theta}(x_t, t)$ which depends on x_t and t . Okay. This was for the unconditional ddpm. Okay. Now I have written the first term of the equation like this. Okay.

[23:57] So when I rewrite this equation it becomes like this. gradient with respect to $x_t \log p(x_t | y)$ is equal to I have just replaced this term here plus same thing I have put here this one this has come here gradient with respect to $x_t \log p(x_t | y)$ okay so this is the equation right now in my hand this is basically what is your ϵ_{θ} it is the old noise that is predicted by your DDPM. Okay, without any condition and there is an addition of a new term here which is the gradient term that is coming from the classifier.

[24:46] Now because of the addition of this term here, it gives a new noise which is oriented towards a particular class. Okay. So it gives a new noise so that when you remove that noise from the image the generated image becomes oriented towards the particular class Y . Okay, which is the dog class here. I hope this is clear till now. Okay. So from here what do we do? I have just written that expression once again and what I have done is I have written the previous expression okay $\log p(x_t | y)$ uh you know plus this one now this first this the first term here the left hand side term I am assuming that this is written as $-\frac{1}{\sqrt{1 - \alpha_t}}$ by square root of $1 - \alpha_t$ $\epsilon_{\theta}(x_t, t)$.

[25:51] What is this ϵ_{θ} ? This is the noise new noise that is predicted taking into consideration the classifier gradient. I am expecting the new noise ϵ_{θ} you know which depends on x_t and y . So this y is the conditioning term that is coming here inside my new noise prediction. I am replacing this left hand side with this expression. I'm assuming that this will equate when I put this my equation will be satisfied. So this is my left hand side term. This is once again I have seen what it was in my previous slide itself. $\log p(x_t | y)$ we can write it for t also and when I write this one what happens is if I replace these terms here and for this term I replace here and this term then I

[26:58] multiply both the sides with minus square root of $1 - \alpha_t$ I get this expression you can do it okay and you will see you are getting this expression. So what is this expression giving? This expression is nothing which is giving your ϵ_{θ}

hat. ϵ_{hat} is the new noise that is predicted using the classifier guidance. Okay. And what if you see there is a term here which is the guidance scale as I was telling in my previous uh slides. Okay.

[27:34] So what exactly this guidance scale does? It is basically scaling how much of your gradient how much strength you will give to the gradient term that is coming from the classifier and then you put it in your normal model diffusion model. Okay. And we have seen if it is too high or too low what are the consequences of that one. Okay. Diversity versus precision. So that becomes a challenge there. Okay. So this S is called as the guidance scale. And we have got this is the new noise predicted for uh your classifier guided diffusion model. Okay.

[28:13] The new noise is oriented towards a particular class. Okay. And this is basically your classifier guided noise prediction equation. Okay. So this is your classifier guided noise prediction equation. And we know as I was telling for small s the images will be diverse but the conditioning will be weak. That means images may not look like a dog's image. For large s values it is strong conditioning but the diversity in the images generated will be less. Okay. So that's the trade-off that we have to do in finding an optimal value of the guidance scale.

[28:58] Now we come to the sampling equation. The guided noise estimate replaces the standard noise estimate in the reverse DDPM's reverse update. So we have the samp we know the sampling equation for normal DDPM. Now in that sampling equation what we do is the guided noise we will replace the standard noise estimate. We will replace standard noise estimate with the guided noise estimate. That's all. And this is the equation for sampling x_{t-1} given you have x_t value. So square root of you know α_{t-1} within bracket x_t minus square root of $1 - \alpha_t$ bar square root of α_t bar into ϵ_{hat} plus square root of $1 - \alpha_{t-1}$ bar into ϵ . Okay. So this is the sampling equation that you get for

[29:57] the classifier guided diffusion models. And this image I have taken from the original paper where we have images classifier guidance on the class Pembroke Welsh Corgi and we have seen what is the images that are produced with $S=1$ and S equal to 100. Okay. So sorry this is s equal to 10. Okay. And if you see the left hand side it doesn't produce very convincing results with a smaller s value and the right hand side we with a higher s value we get consistent images of that particular class. Okay. And this I have taken from the original paper by by Dhari Atal which was um presented in Europe's 2021 conference. Okay.

[31:02] Now what are the limitations of classifier guided diffusion model? Let us have that one. The first and foremost limitation is it requires training an additional classifier and that to specifically on noisy images X_T . So together with your normal architecture for the diffusion models, you have to train another classifier specifically on noisy images provided the necessary gradients at its steps of the diffusion process

to provide the necessary gradient. Okay. So that is important that is a limitation. You are required to have two trainings. one is for the normal DDPM unconditional one and another classifier one. The second limitation is you cannot use off-the-shelf classifiers like ResNet

[31:59] which is trained on image data set because it will not work properly as the off-the-shelf um classifiers trained on image data set. They are trained on clean images not on noisy images. Okay. Secondly, that is the second uh limitation. Then complex pipeline as two separate networks as I just told. Then classifier gradients must be compute at every time step t increasing the computational cost during sampling. Okay. So computational cost increases.

[32:36] Okay. Also there is something called as the adversarial vulnerability. Diffusion model sometimes creates artifacts which looks very similar to target class but they don't look human they look very artificial so this is called as adversarial vulnerability so there are limitations of classifier guided diffusion models and with this one I come to the end of today's session before I conclude let us quickly summarize so in the In today's lecture, we have seen what is the motivation for classifier guided diffusion models.

[33:18] Basically, if I need to synthesize images that belong to a particular class, then I need some guidance mechanism, which is the concept behind classifier guided diffusion models. And in this classifier guided diffusion model, there are two components. One is the normal diffusion model and another one is the classifier. Now, the classifier is trained on noisy images. to predict a particular class. Okay. And the gradient of this classifier output is given to the normal uh DDPM architecture so that the image generation can be steered towards that particular class. That is the core idea of this one. And then we have seen some mathematical steps how we can formulate that mathematically. What is the

[34:10] expression for the noise term in case of classifier guided diffusion? Because that noise will not be same as your original or previously predicted noise by the DDPM because what happens here is you have an another term which is coming from the classifier. And that has led us to give another new um noise term which is $\epsilon_{\hat{\theta}}$. And that will help in generating images that that amount of noise when taken out from the uh noisy image will help in steering the image generation towards a particular class. Okay. And we have seen that there is a term which is called as the guidance scale. it determines how much strength you want to give to that um classifier guidance. Okay. If it is very low then it will uh the images

[35:05] generated will not look very uh uh very close to a particular class say a dog class. If it is too high then it will look like a dog but at the same time the diversity of the images that are produced reduces. Okay. So these are the limit these are the trade-offs that we have to uh sort with and then finally we have seen what is the

sampling equation uh and how the classifier guided uh diffusion images uh look like with a lower value of uh uh guidance uh strength and guidance scale and higher value of guidance scale. And finally we have seen what are the limitations of this classifier guidance uh diff guided diffusion model. One is you require a extra classifier that needs to be trained in noisy images. You cannot use

[35:54] off-the-shelf classifiers like ResNet which are which are trained on clean images that is not uh that will not work because here we are dealing with noisy images and also the complexity of the pipeline increases because you have a unit for denoising plus you have a classifier for uh you know giving the gradient. Okay. So these are the uh limitations. With this one I come to the conclusion of today's session. In my next session we will discuss about classifier free diffusion models.

[36:31] That is one step ahead that can we use or can we have uh conditioned image generation without using a classifier. explicitly without using a classifier. Can we do that one? And that is the topic of my next uh session classifier free guidance in diffusion models. Okay. So with this one I come to the end of today's session. I hope you all had got an understanding of how um classifier uh based diffusion models work. Okay. And till we meet for the next session with classifier free guidance I urge all of you to go through the original paper.

[37:17] You will get lot more details about the model its implementation the mathematical derivations there. And till we meet for the next uh session keep learning happy learning. Thank you all.